

Tushar Chaurasia

AI/ML Engineer | Deep Learning | Systems Programming | C++

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Computer Science undergraduate focused on **high-performance systems** and **AI/ML**, with experience in **C++**, **deep learning**, **compilers**, **memory-safe languages**, **build systems**, and a **quantum circuit simulator**. Strong in **algorithms**, **data structures**, **low-level memory management**, and **scalable system design**. Open-source and stackoverflow.com contributor (3K+ reputation), with solutions impacting 71K+ developers.

SKILLS

- **Core / Primary:**
C++, C, Python, Data structures & algorithms, Systems programming, Memory management & low-level optimization, Performance-oriented programming
- **AI / Machine Learning:**
Regression, Classification, Transfer Learning, Neural Network, Deep Learning, CNN, Neural Networks, Model training, Tensorflow, Scikit-learn, Pandas, Numpy, NLP foundations
- **Development:**
Node.js, React.js, REST APIs, HTML, CSS, Javascript, Vite
- **Foundational Concepts:**
Quantum computing (gate-based simulation), Operating systems concepts, Linear algebra, Probability, Statistics, Software engineering, Agile methodologies
- **Tools & Environment:**
Git, GitHub, Linux, VS Code, Visual Studio, Jupyter Notebook, JetBrains Tools, Valgrind
Databases: MySQL, MongoDB, NoSQL
- **Soft Skills:**
Analytical problem solving, Debugging, Technical communication, Leadership in technical teams.

PROJECTS

QUBITVERSE

C++, Node.js, vite, React.js, recharts, and React-Konva

- Led a **group of 14+** individual developers across the nation to build a **gate-based quantum circuit simulator** using a high-performance C++ backend with an interactive React GUI.
- Implemented **state-vector representation** to simulate arbitrary **n-qubit systems**, supporting **17+ quantum gates** with correct unitary transformations.
- Focused on **numerical correctness, performance, and scalability**.
- Enabled step-by-step visualization of quantum state evolution for **circuit debugging** and learning.

QUANTZ

C++, Python, Flask, Node.js, vite, React.js, and ReactFlow

- Developed a platform for **designing, testing, and backtesting trading strategies** using a **SIMD (AVX/SSE) powered C++ computation core**, for CPU-based parallel processing and efficiency.
- Implemented **technical indicators, reusable strategy logic, and data-driven analysis pipelines**.
- Bridged **mathematical models with practical trading workflows**, emphasizing modularity and extensibility.
- Designed architecture to support **future strategy expansion and performance-critical computations**.

MEDCHECK

Python, Tensorflow, MobileNetV2, OpenCV2, Flask, Node.js, and React.js

- Developed a MobileNetV2-based CNN (transfer learning) to classify genuine vs. counterfeit Pfizer medicines, achieving **85% accuracy on 293 test images**.
- Delivered **F1-scores of 0.88 (genuine) and 0.79 (counterfeit)**, with **0.87 recall for counterfeit**, emphasizing reliable fraud detection.
- Used **data augmentation** and fine-tuning to improve robustness on real-world packaging variations.

RUNCPP

C++, Node.js, and Electron.js

- Built a cross-platform build system with a **custom lexer and parser** for **declarative build configuration**.
- Implemented **dependency resolution, target execution**, and incremental builds.
- Developed an **Electron-based GUI** for build visualization and project control.
- Focused on improving build **automation**, clarity, and maintainability for **large projects**.

HORIZON

C++

- Designed a **memory-safe systems programming language** combining C++-style syntax with **compiler-enforced ownership rules**.
- Implemented **heap-managed types, move semantics, and temporary borrowing (ref)**.
- Eliminated **dangling pointers, double frees, and invalid aliasing** at compile time.
- Focused on providing **low-level control with compile-time safety guarantees**.

EXPERIENCE

Open Source Connect

Project Admin

📍 Remote

📅 Aug 2025 – Sept 2025

- Managed repositories, issue tracking, and contributor workflows to keep open-source projects organized and active.
- Reviewed contributions, enforced contribution guidelines, and coordinated between maintainers and contributors.
- Planned timelines, tracked progress, and ensured consistent project delivery and community engagement.

VERGE

Event Coordinator

📍 Sonapat, Delhi-NCR

📅 Jan 2025 – Feb 2025

- Planned and executed technical events, managing schedules, logistics, and on-ground operations.
- Coordinated with speakers, judges, sponsors, and student teams to ensure smooth event flow.
- Led volunteer teams, handled real-time problem-solving, and ensured events ran on time and within scope.

WEBBYTES

DSA/CP Coordinator

📍 Sonapat, Delhi-NCR

📅 Apr 2024 – Aug 2025

- Organized multiple workshops for 100+ students on-campus to encourage coding and DSA culture in the college.
- Taken lectures for second- and third-year Computer science Engineering students on string, linked list, vector, and multiple data structures in the workshops.

EDUCATION

Pursuing B.Tech. in Computer Science and Engineering with Specialization in Artificial Intelligence and Data Science

SRM University Delhi-NCR, Sonapat – *SGPA: 8.739, CGPA: 8.19*

📅 June 2023 – June 2027

CERTIFICATES & ACHIEVEMENTS

- An appreciation and letter of recommendation from Mr. Sandeep Jain, CEO of GeeksforGeeks, acknowledging my contribution & winner in the Bi-Wizard Coding Contest 2022. ([Link](#))
- Finalist (Top 15 teams) at Hackstacy Hackathon 2025, with final round hosted at Microsoft, Noida. ([Link](#))
- Project Admin at Open-Source Connect India 2025 for my project QubitVerse. ([Link](#))
- Supervised Machine Learning: Regression and Classification by DeepLearning.AI and Stanford Online. ([Link](#))
- Advanced Learning Algorithms by DeepLearning.AI and Stanford Online. ([Link](#))
- Unsupervised Learning, Recommenders, Reinforcement Learning by DeepLearning.AI and Stanford Online. ([Link](#))
- Machine Learning Specialization by DeepLearning.AI and Stanford Online with instructor Andrew Ng. ([Link](#))